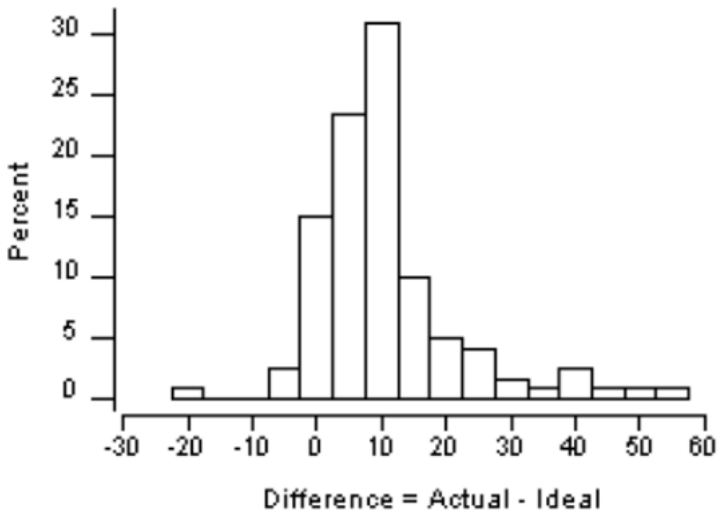
Circle the correct answer for each question. If you're unsure, you can write numbers 1, 2, 3, etc. next to the choices to get partial credit.

Question permutation table:

Version		1	2	3	4
Question					
1	→	9	1	2	8
2*	→	10	2	3	9
3*	→	11	3	4	10
4	→	3	10	6	11
5*	→	4	11	7	12
6	→	12	9	5	1
7	→	2	8	12	3
8	→	5	4	8	4
9*	→	6	5	9	5
10*	→	7	6	10	6
11*	→	8	7	11	7
12	→	1	12	1	2

*: Question follows previous

The following histogram shows the distribution of the difference between the actual and "ideal" weights for 119 female students. Ideal weights are responses to the question "What is your ideal weight"? The difference = actual – ideal.



1. What is the approximate shape of the distribution?

   (a) Extremely skewed right
   (b) Skewed left
   (c) Approximately normal with slight skew to the right
   (d) Unable to determine

   Correct answers: (c)

Answer permutations: (All answers permuted)

Version		1	2	3	4
Answer					
(a)	→	(b)	(c)	(a)	(c)
(b)	→	(a)	(a)	(d)	(a)
(c)	→	(d)	(b)	(b)	(d)
(d)	→	(c)	(d)	(c)	(b)

2. How does the mean compare to the median in this data set?

   (a) The mean is larger than the median.
   (b) The median is larger than the mean.
   (c) The median and mean are approximately equal.
   (d) The mean and the median cannot be compared without calculations.

Correct answers: (a)

Answer permutations: (All answers permuted)

Version		1	2	3	4
Answer					
(a)	→	(c)	(b)	(c)	(d)
(b)	→	(a)	(c)	(d)	(c)
(c)	→	(b)	(a)	(b)	(b)
(d)	→	(d)	(d)	(a)	(a)

3. If the observation at -20 pounds was removed from the data set, the mean _____.

   (a) would increase
   (b) would decrease
   (c) would remain the same
   (d) may or may not change

Correct answers: (a)

Answer permutations: (All answers permuted)

Version		1	2	3	4
Answer					
(a)	→	(c)	(d)	(d)	(d)
(b)	→	(d)	(c)	(a)	(b)
(c)	→	(a)	(a)	(c)	(a)
(d)	→	(b)	(b)	(b)	(c)

A hot dog manufacturer asserts samples 10 hot dogs and calculates the average fat content to be 18 grams per hot dog and a standard deviation of 1.1 grams. We know from past studies that fat contents in hot dogs from this manufacturer follow an approximately normal distribution.

4. The value 18 is the _____.

   (a) sample size
   (b) statistic
   (c) parameter
   (d) standard deviation

Correct answers: (b)

Answer permutations: (All answers permuted)

Version		1	2	3	4
Answer					
(a)	→	(a)	(c)	(a)	(a)
(b)	→	(b)	(a)	(c)	(d)
(c)	→	(d)	(b)	(b)	(b)
(d)	→	(c)	(d)	(d)	(c)

5. Is the sampling distribution of average fat content of these hot dogs approximately normal?

   (a) No, since the sample size is too small for the Central Limit Theorem to hold.
   (b) No, since the manufacturer does not know if the population of hot dogs is more than 20 times the size of the sample.
   (c) Yes, since the population of fat contents in hot dogs is approximately normal.
   (d) Not enough information exists to determine if the sampling distribution of x is approximately normal.

   Correct answers: (c)

   Answer permutations: (All answers permuted)

Version		1	2	3	4
Answer					
(a)	→	(b)	(a)	(b)	(a)
(b)	→	(d)	(b)	(a)	(b)
(c)	→	(a)	(c)	(c)	(c)
(d)	→	(c)	(d)	(d)	(d)

6. Which of the following distributions would have the smallest standard deviation?

   (a) Number of miles put on a rental car in one day
   (b) Average number of miles put on a rental car per day across 5 customers
   (c) Average number of miles put on a rental car per day across 25 customers
   (d) Distributions of all of the above would have the same standard deviation.

   Correct answers: (c)

   Answer permutations: (All answers permuted)

Version		1	2	3	4
Answer					
(a)	→	(c)	(c)	(d)	(a)
(b)	→	(b)	(a)	(c)	(b)
(c)	→	(d)	(d)	(a)	(c)
(d)	→	(a)	(b)	(b)	(d)

7. Most experts recommend college students get at least 8 hours of sleep per night. A random sample is collected of 112 students at a college by asking how many hours of sleep each student got last night. The sample mean amount of sleep last night was 7.6 hours, with a sample standard deviation of 2.1 hours. The sample distribution of hours of sleep was slightly skewed right.
   Use the sample results to estimate the standard error of the sampling distribution of sample means.

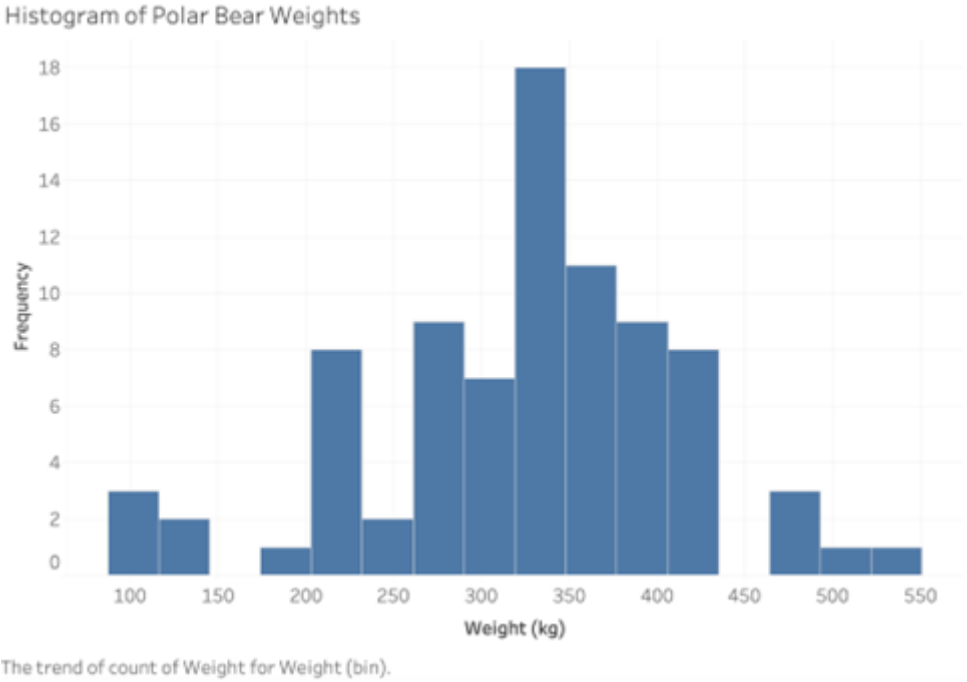
   (a) 2.1
   (b) 0.198
   (c) 7.6
   (d) 0.718

   Correct answers: (b)

   Answer permutations: (All answers permuted)

Version		1	2	3	4
Answer					
(a)	→	(c)	(b)	(b)	(c)
(b)	→	(d)	(a)	(a)	(d)
(c)	→	(a)	(d)	(d)	(a)
(d)	→	(b)	(c)	(c)	(b)

Adult male polar bears are expected to weigh, on average, 370 kg, but there is concern by some scientists that because of recent warmer winters that polar bears might be struggling to gain weight. The US Geological Survey (USGS) conducted a study in the Southern Beaufort Sea on the weight of polar bears. Eighty-three adult male polar bears were captured between March and May of the years 1990 and 2006 and their weights were recorded. The sample mean weight was 324.6 kg, and the sample standard deviation was 88.3 kg.
A histogram of the 83 polar bear weights in the sample is shown below.



8. Are the validity conditions for theory-based inference met for these data?

   (a) Yes, since at least 10 polar bears weighed more than 370 kg and at least 10 weighed less than 370 kg.
   (b) Yes, since the sample size is larger than 20 and the sample distribution is not strongly skewed.
   (c) No, since the sample distribution is slightly skewed left.
   (d) Unable to determine.

   Correct answers: (b)

   Answer permutations: (All answers permuted)

Version		1	2	3	4
Answer					
(a)	→	(a)	(c)	(b)	(c)
(b)	→	(c)	(b)	(d)	(a)
(c)	→	(d)	(a)	(c)	(b)
(d)	→	(b)	(d)	(a)	(d)

9. What are the appropriate null and alternative hypotheses?

   (a) $H_0 : \overline{x} = 370$ versus $H_A : \overline{x} \neq 370$
   (b) $H_0 : \overline{x} = 370$ versus $H_A : \overline{x} < 370$
   (c) $H_0 : \mu = 370$ versus $H_A : \mu \neq 370$
   (d) $H_0 : \mu = 370$ versus $H_A : \mu < 370$

   Correct answers: (c)

Answer permutations: (All answers permuted)

	Version	1	2	3	4
Answer					
(a)	→	(d)	(a)	(b)	(b)
(b)	→	(c)	(b)	(a)	(d)
(c)	→	(b)	(d)	(d)	(c)
(d)	→	(a)	(c)	(c)	(a)

10. The sample mean of 324.6 kg lies _____(1)_____ standard errors _____(2)_____ the expected average polar bear weight of 370 kg.

    (a) 4.68, above
    (b) 4.68, below
    (c) 0.51, above
    (d) 0.51, below

    Correct answers: (b)

    Answer permutations: (All answers permuted)

	Version	1	2	3	4
Answer					
(a)	→	(a)	(d)	(c)	(d)
(b)	→	(d)	(c)	(d)	(c)
(c)	→	(b)	(a)	(a)	(a)
(d)	→	(c)	(b)	(b)	(b)

11. A theory-based inference applet gives a p-value less than 0.00001. Do these results provide evidence that the true mean weight of polar bears is less than 370 kg?

    (a) No, since the sample results could have happened by chance.
    (b) Yes, since 324.6 kg is less than 370 kg.
    (c) Yes, since the p-value is small.
    (d) No, since the p-value is large.

    Correct answers: (c)

    Answer permutations: (All answers permuted)

	Version	1	2	3	4
Answer					
(a)	→	(a)	(d)	(a)	(a)
(b)	→	(c)	(a)	(c)	(c)
(c)	→	(d)	(b)	(d)	(b)
(d)	→	(b)	(c)	(b)	(d)

12. If the p-value is very small, which of the following is true?

    (a) Very strong evidence exists against the null hypothesis.
    (b) Very strong evidence exists against the alternative hypothesis.
    (c) The null hypothesis has been proven.
    (d) The alternative hypothesis has been proven.

    Correct answers: (a)

    Answer permutations: (All answers permuted)

	Version	1	2	3	4
Answer					
(a)	→	(a)	(d)	(c)	(a)
(b)	→	(d)	(a)	(a)	(b)
(c)	→	(b)	(c)	(d)	(c)
(d)	→	(c)	(b)	(b)	(d)